

Install Java

```
sudo apt update  
sudo apt install openjdk-8-jdk -y
```

Check Java:

```
java -version
```

Create Hadoop User (if not already created)

```
sudo adduser hduser  
  
sudo apt install ssh -y  
  
su - hduser
```

Install SSH

```
sudo apt install ssh -y
```

Generate SSH key:

```
ssh-keygen -t rsa -P ""
```

Enable SSH login:

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys  
chmod 0600 ~/.ssh/authorized_keys
```

ssh localhost (Note this command works without asking password if it asks password the process is incorrect)

Download Hadoop

Example (version you used):

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
```

Extract:

```
tar -xvzf hadoop-3.3.6.tar.gz  
mv hadoop-3.3.6 hadoop
```

Configure Environment Variables

Open .bashrc:

```
nano ~/.bashrc
```

Add:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

```
export HADOOP_HOME=/home/hduser/hadoop
```

```
export PATH=$PATH:$HADOOP_HOME/bin
```

```
export PATH=$PATH:$HADOOP_HOME/sbin
```

```
export HADOOP_MAPRED_HOME=$HADOOP_HOME
```

```
export HADOOP_COMMON_HOME=$HADOOP_HOME
```

```
export HADOOP_HDFS_HOME=$HADOOP_HOME
```

```
export YARN_HOME=$HADOOP_HOME
```

```
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
```

```
source ~/.bashrc
```

Configure hadoop-env.sh

```
nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh
```

Add:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

Configure core-site.xml

```
nano $HADOOP_HOME/etc/hadoop/core-site.xml
```

Add inside <configuration>:

```
<property>
```

```
<name>fs.defaultFS</name>
```

```
<value>hdfs://localhost:9000</value>
```

```
</property>
```

Configure hdfs-site.xml

```
nano $HADOOP_HOME/etc/hadoop/hdfs-site.xml
```

Add:

```
<property>
```

```
<name>dfs.replication</name>
```

```
<value>1</value>
```

```
</property>
```

Configure mapred-site.xml

```
nano $HADOOP_HOME/etc/hadoop/mapred-site.xml
```

Add:

```
<configuration>
```

```
<property>
```

```
<name>mapreduce.framework.name</name>
```

```
<value>yarn</value>
```

```
</property>
```

```
<property>
```

```
<name>yarn.app.mapreduce.am.env</name>
```

```
<value>HADOOP_MAPRED_HOME=/home/hduser/hadoop</value>
```

```
</property>
```

```
<property>
```

```
<name>mapreduce.map.env</name>
```

```
<value>HADOOP_MAPRED_HOME=/home/hduser/hadoop</value>
```

```
</property>
```

```
<property>
```

```
<name>mapreduce.reduce.env</name>
```

```
<value>HADOOP_MAPRED_HOME=/home/hduser/hadoop</value>
```

```
</property>
```

```
</configuration>
```

Configure yarn-site.xml

```
nano $HADOOP_HOME/etc/hadoop/yarn-site.xml
```

Add:

```
<property>
```

```
<name>yarn.nodemanager.aux-services</name>
```

```
<value>mapreduce_shuffle</value>
```

```
</property>
```

```
hdfs namenode -format
```

start-dfs.sh

start-yarn.sh

jps

hdfs dfsadmin -safemode leave

hdfs dfs -mkdir -p /user/hduser/input

nano file.txt

(write some words)

hdfs dfs -put file.txt /user/hduser/input

hadoop jar \$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar wordcount /user/hduser/input /user/hduser/output

hdfs dfs -cat /user/hduser/output/part-r-00000

stop-yarn.sh

stop-dfs.sh